SIGNALS AND SYSTEMS - EXPERIMENT 2

TASK1

Input Coding

```
clc;
clear;
close all;

%% Index
n = -10:10;

%% Input signal
x = sin(0.3*pi*n);

%% System Output
y = zeros(size(n));

for k = 1:length(n)
    idx = find(n == 2*n(k));
    if ~isempty(idx)
        y(k) = x(idx);
    end
end

%% Plot
figure;
subplot(2,1,1);
stem(n,x,'filled');
title('Input Signal x[n]');
xlabel('n');
ylabel('Amplitude');
grid on;
```

```
subplot(2,1,2);
stem(n,y,'filled');
title('Output Signal y[n]=x[2n]');
xlabel('n');
ylabel('Amplitude');
grid on;

%% Linearity Test
x1 = sin(0.2*pi*n);
x2 = cos(0.4*pi*n);

a = 2;
b = 3;

x_comb = a*x1 + b*x2;

Tx_comb = zeros(size(n));
Tx1 = zeros(size(n));
Tx2 = zeros(size(n));

for k=1:length(n)
   idx=find(n==2*n(k));
   if ~isempty(idx)
      Tx_comb(k)=x_comb(idx);
      Tx1(k)=x1(idx);
      Tx2(k)=x2(idx);
   end
end

Linearity_Error = max(abs(Tx_comb-(a*Tx1+b*Tx2)));

fprintf('Linearity Error = %f\n',Linearity_Error);
```

```
%% Time Invariance Test
n0 = 2;

x_delay = sin(0.3*pi*(n-n0));

Tx_delay = zeros(size(n));
for k=1:length(n)
   idx=find(n==2*n(k));
   if ~isempty(idx)
      Tx_delay(k)=x_delay(idx);
   end
end

y_delay = zeros(size(n));
for k=1:length(n)
   idx=find(n==n(k)-n0);
   if ~isempty(idx)
      y_delay(k)=y(idx);
   end
end

difference = Tx_delay - y_delay;

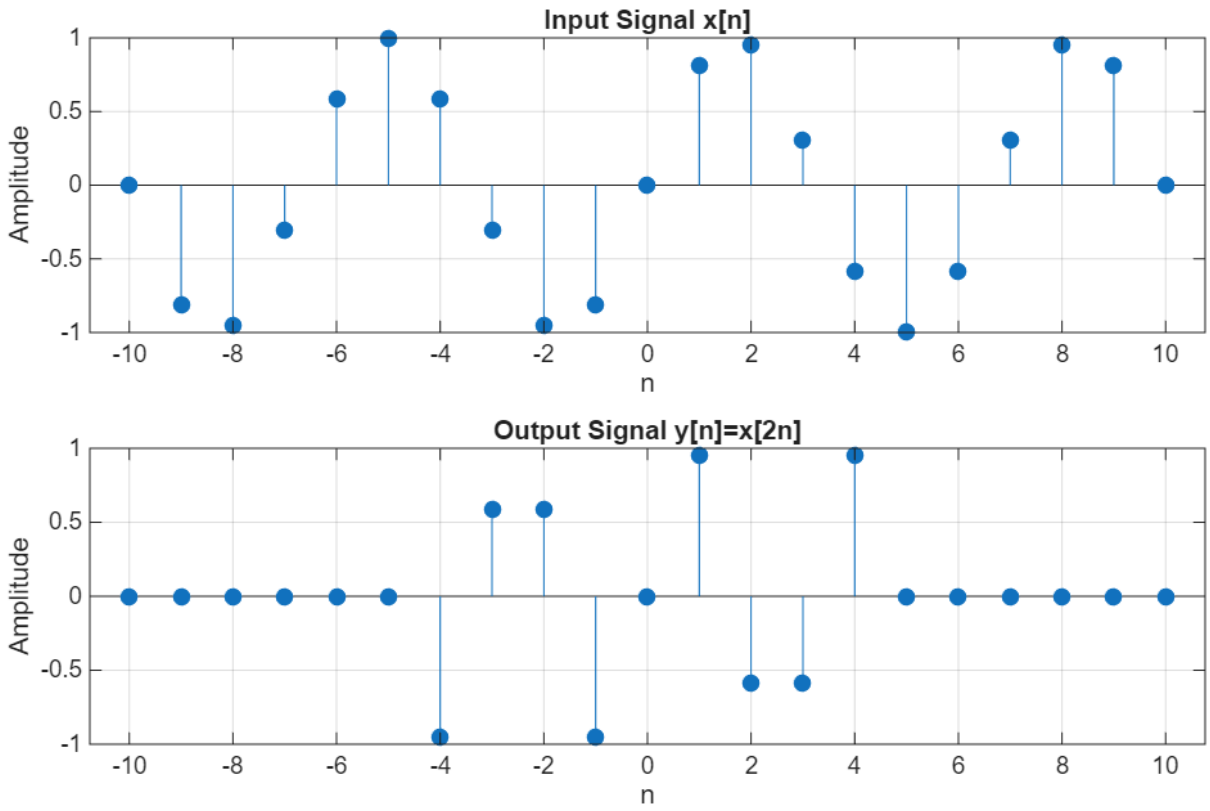
disp('Difference between outputs:')
disp(difference)

idx_diff = find(abs(difference)>1e-6);

fprintf('Indices where outputs differ:\n');
disp(n(idx_diff))
```

```
%% Classification
disp('Time Scaling System')
disp('Linear       : YES')
disp('Time Invariant: NO')
disp('Causal       : NO')
disp('Memory       : YES')
disp('Stable       : YES')
```

Output



TASK TWO

INPUT CODING

```
clc;
clear;
```

```
close all;

n = -5:5;
x = sin(pi*n/4);

% Output
y = sin(-pi*n/4);

figure;
subplot(2,1,1);
stem(n,x,'filled');
title('Input x[n]');
grid on;

subplot(2,1,2);
stem(n,y,'filled');
title('Output y[n]=x[-n]');
grid on;

%% Time Invariance Test
n0 = 2;

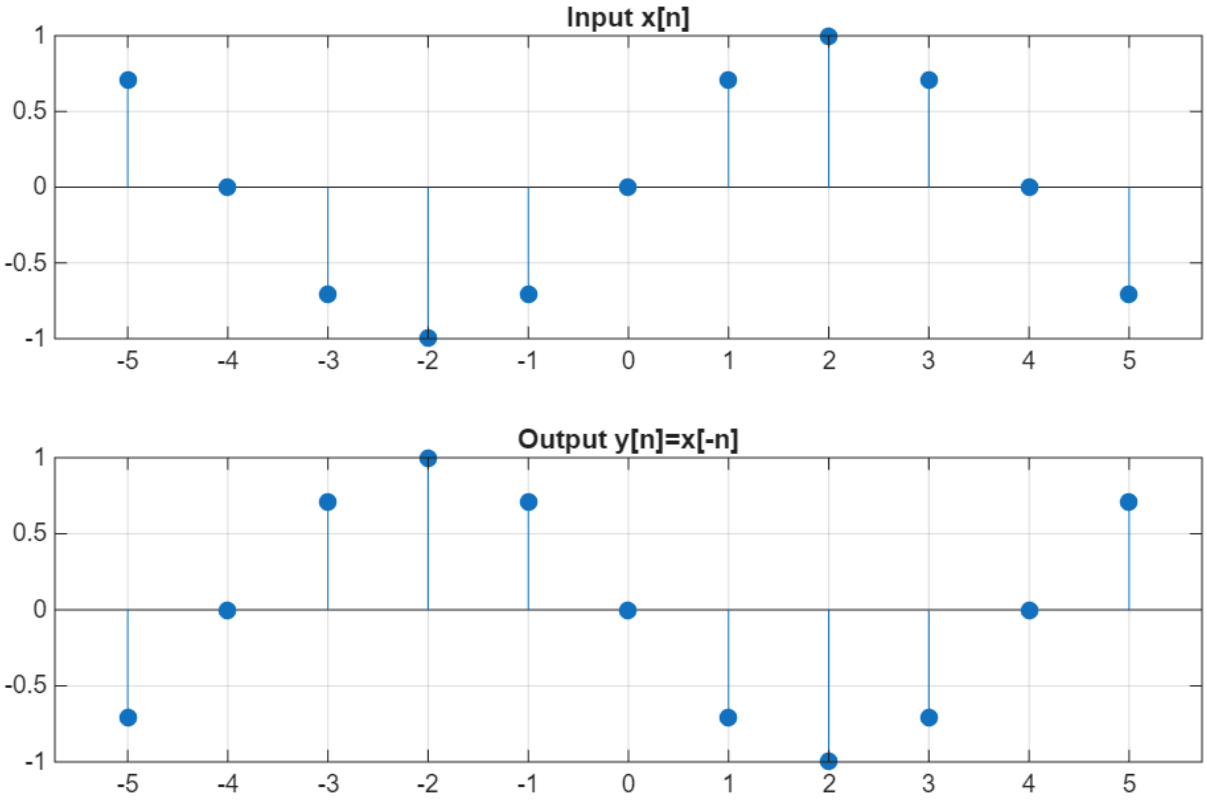
Tx1 = sin(-pi*(n-n0)/4);   % T{x[n-2]}
Ty1 = sin(-pi*n/4-pi*n0/4); % T{x[n]} delayed

disp('Difference = ');
disp(Tx1-Ty1);

disp('System Properties');
disp('Linear : YES');
disp('Time Invariant : NO');
```

```
disp('Causal : NO');
disp('Memory : YES');
disp('Stable : YES');
```

OUTPUT



TASK 3

INPUT CODING

```
clc;
clear;
close all;

n = -5:5;
x = sin(pi*n/4);

% Output
y = n.*x;
```

```
figure;
subplot(2,1,1);
stem(n,x,'filled');
title('Input x[n]');
grid on;

subplot(2,1,2);
stem(n,y,'filled');
title('Output y[n]=n*x[n]');
grid on;

%% Time Invariance Test (n0 = 1)

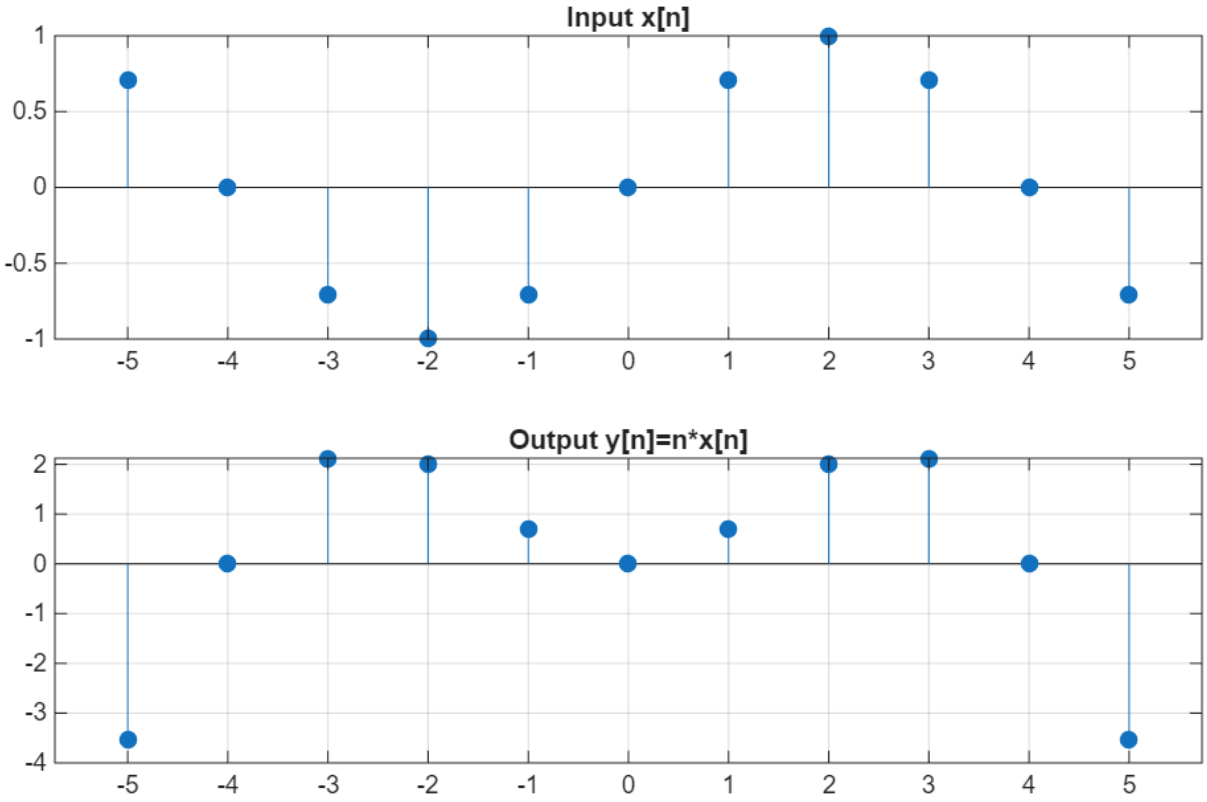
n0 = 1;

Tx1 = n.*sin(pi*(n-n0)/4);       % T{x[n-1]}
Ty1 = (n-n0).*sin(pi*(n-n0)/4);  % T{x[n]} delayed

disp('Difference = ');
disp(Tx1-Ty1);

disp('System Properties');
disp('Linear : YES');
disp('Time Invariant : NO');
disp('Causal : YES');
disp('Memory : NO');
disp('Stable : NO');
```

OUTPUT


Input x[n]
1
0.5
0
-0.5
-1
-5
-4
-3
-2
-1
0
1
2
3
4
5
Output y[n]=n*x[n]
2
1
0
-1
-2
-3
-4
-5
-4
-3
-2
-1
0
1
2
3
4
5